

FC7xxx GPT Integration Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

Table of Contents

Revision History.....	2
Table of Contents	3
Chapter 1 Introduction	4
1.1 Introduction	4
Chapter 2 Building	5
2.1 Dependencies on Other Modules.....	5
2.2 Files Required for Compile	5
2.3 Add Plug-ins	7
Chapter 3 Memory	8
3.1 Sections in Memory Map	8
Chapter 4 Exclusive Area.....	10
Chapter 5 Interrupt Service Routine (ISR).....	12
Chapter 6 Error Report.....	13
6.1 Det.....	13
6.2 Dem.....	14
Chapter 7 Function Calls.....	15
7.1 Function Calls during Startup	15
7.2 Function Calls during Shutdown	15
7.3 Function Calls during Wake-up.....	15
7.4 Function Calls during Runtime	15
Chapter 8 Other Requirements	16
8.1 Notification, Callback, Callout	16
8.2 Macros	16
Chapter 9 Integration Steps.....	17

Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for the GPT module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Mcu: This module provides the clock reference point for GPT module
- Common: This module is the basic module which used to choose the chip.

Module build dependency

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from interrupts (Exclusive Areas).
- Det: This module is necessary for enabling Development error detection.

Module initialization dependency

- Mcu: To use the GPT module, the user needs to initialize the MCU module first

2.2 Files Required for Compile

GPT module files:

- _MCAL/Src/Gpt/Src/Gpt.c
- _MCAL/Src/Gpt/Src/Gpt_Fcpit.c
- _MCAL/Src/Gpt/Src/Gpt_Tstmp.c
- _MCAL/Src/Gpt/Src/Gpt_Aontimer.c
- _MCAL/Src/Gpt/Src/Gpt_Ftu.c
- _MCAL/Src/Gpt/Src/Gpt_Hw.c
- _MCAL/Src/Gpt/Src/Gpt_Rtc.c
- _MCAL/Src/Gpt/include/Gpt.h
- _MCAL/Src/Gpt/include/Gpt_Aontimer.h
- _MCAL/Src/Gpt/include/Gpt_Aontimer_RegOps.h
- _MCAL/Src/Gpt/include/Gpt_Aontimer_Types.h
- _MCAL/Src/Gpt/include/Gpt_Ftu.h
- _MCAL/Src/Gpt/include/Gpt_Ftu_RegOps.h
- _MCAL/Src/Gpt/include/Gpt_Ftu_Types.h
- _MCAL/Src/Gpt/include/Gpt_Hw.h
- _MCAL/Src/Gpt/include/Gpt_Hw_Irq.h
- _MCAL/Src/Gpt/include/Gpt_Irq.h
- _MCAL/Src/Gpt/include/Gpt_MemMap.h
- _MCAL/Src/Gpt/include/Gpt_Rtc.h
- _MCAL/Src/Gpt/include/Gpt_Rtc_Reg.h
- _MCAL/Src/Gpt/include/Gpt_Rtc_RegOps.h
- _MCAL/Src/Gpt/include/Gpt_Rtc_Types.h
- _MCAL/Src/Gpt/include/Gpt_Fcpit.h
- _MCAL/Src/Gpt/include/Gpt_Fcpit_RegOps.h

- _MCAL/Src/Gpt/include/Gpt_Fcpit_Types.h
- _MCAL/Src/Gpt/include/Gpt_Tstmp.h
- _MCAL/Src/Gpt/include/Gpt_Tstmp_RegOps.h
- _MCAL/Src/Gpt/include/Gpt_Tstmp_Types.h
- _MCAL/Src/Gpt/include/Gpt_version.h
- _MCAL_generate/src/Gpt_PBcfg.c
- _MCAL_generate/src/Gpt_Cfg.c
- _MCAL_generate/include/Gpt_Cfg.h

Common module files:

- _MCAL/Src/Common/include/Aontimer_Reg.h
- _MCAL/Src/Common/include/arm_cortex_asm.h
- _MCAL/Src/Common/include/Common_MemMap.h
- _MCAL/Src/Common/include/Compiler_Cfg.h
- _MCAL/Src/Common/include/CompilerDefinition.h
- _MCAL/Src/Common/include/Cpm_Reg.h
- _MCAL/Src/Common/include/Eth_GeneralTypes.h
- _MCAL/Src/Common/include/Fcpit_Reg.h
- _MCAL/Src/Common/include/Ftu_Common.h
- _MCAL/Src/Common/include/Ftu_Reg.h
- _MCAL/Src/Common/include/Ftu_RegOps.h
- _MCAL/Src/Common/include/Gpio_Reg.h
- _MCAL/Src/Common/include/IRQRouter.h
- _MCAL/Src/Common/include/Mb_Reg.h
- _MCAL/Src/Common/include/Mb_RegOps.h
- _MCAL/Src/Common/include/Mcal.h
- _MCAL/Src/Common/include/Platform_Types.h
- _MCAL/Src/Common/include/Port_Reg.h
- _MCAL/Src/Common/include/Scm_Reg.h
- _MCAL/Src/Common/include/Scm_RegOps.h
- _MCAL/Src/Common/include/SpinLock.h
- _MCAL/Src/Common/include/StdRegMacros.h
- _MCAL/Src/Common/include/Tstmp_Reg.h
- _MCAL/Src/Common/src/Aontimer_Common.c
- _MCAL/Src/Common/src/Ftu_Common.c
- _MCAL/Src/Common/src/IRQRouter.c
- _MCAL/Src/Common/src/Port_Common.c
- _MCAL/Src/Common/src/SpinLock.c

Det module files:

- Det.h

Rte module files:

- SchM_Gpt.h

2.3 Add Plug-ins

GPT module plug-ins are developed for EB tresos Studio, so, to use GPT plug-ins on the EB tresos Studio, the user needs to add the GPT module to the EB plug-ins folder first.

- 1) Copy the GPT module(_MCAL/EB_Plugins/eclipse/plugins/Gpt) folder to EB tresos plug-ins (EB/tresos/plugins/) folder.
- 2) Set the GPT module output location folder for the generated source file and header files.
- 3) Use EB tresos to configure the GPT module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section Name	Section Type	Description
GPT_START_SEC_CONFIG_DATA_8 GPT_STOP_SEC_CONFIG_DATA_8 GPT_START_SEC_CONFIG_DATA_16 GPT_STOP_SEC_CONFIG_DATA_16 GPT_START_SEC_CONFIG_DATA_32 GPT_STOP_SEC_CONFIG_DATA_32	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).
GPT_START_SEC_CONFIG_DATA_UNSPECIFIED GPT_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	Start and stop of Memory Section for Config Data.
GPT_START_SEC_CONST_BOOLEAN GPT_STOP_SEC_CONST_BOOLEAN GPT_START_SEC_CONST_8 GPT_STOP_SEC_CONST_8 GPT_START_SEC_CONST_16 GPT_STOP_SEC_CONST_16 GPT_START_SEC_CONST_32 GPT_STOP_SEC_CONST_32 GPT_START_SEC_CONST_UNSPECIFIED GPT_STOP_SEC_CONST_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only (rodata).
GPT_START_SEC_CODE GPT_STOP_SEC_CODE GPT_START_SEC_RAMCODE GPT_STOP_SEC_RAMCODE GPT_START_SEC_CODE_AC GPT_STOP_SEC_CODE_AC	Code	Start and stop of memory Section for Code (text).
GPT_START_SEC_VAR_NO_INIT_BOOLEAN GPT_STOP_SEC_VAR_NO_INIT_BOOLEAN GPT_START_SEC_VAR_NO_INIT_8 GPT_STOP_SEC_VAR_NO_INIT_8 GPT_START_SEC_VAR_NO_INIT_16 GPT_STOP_SEC_VAR_NO_INIT_16 GPT_START_SEC_VAR_NO_INIT_32 GPT_STOP_SEC_VAR_NO_INIT_32 GPT_START_SEC_VAR_NO_INIT_UNSPECIFIED GPT_STOP_SEC_VAR_NO_INIT_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).
GPT_START_SEC_VAR_INIT_BOOLEAN GPT_STOP_SEC_VAR_INIT_BOOLEAN GPT_START_SEC_VAR_INIT_8 GPT_STOP_SEC_VAR_INIT_8	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by

Section Name	Section Type	Description
GPT_START_SEC_VAR_INIT_16		startup code(data).
GPT_STOP_SEC_VAR_INIT_16		
GPT_START_SEC_VAR_INIT_32		
GPT_STOP_SEC_VAR_INIT_32		
GPT_START_SEC_VAR_INIT_UNSPECIFIED		
GPT_STOP_SEC_VAR_INIT_UNSPECIFIED		

Chapter 4 Exclusive Area

GPT module using the services of Schedule Manger (SchM) for entering and exiting critical regions.

The following critical regions are used in the GPT driver:

- Gpt_Aontimer.c:
 - Gpt_LL_Aontimer_StartTimer : exclusive area 0
 - Gpt_LL_Aontimer_StopTimer : exclusive area 1
 - Gpt_LL_Aontimer_EnableInterrupt : exclusive area 2
 - Gpt_LL_Aontimer_DisableInterrupt : exclusive area 6
 - Gpt_LL_Aontimer_StartPredefTimer : exclusive area 4
 - Gpt_LL_Aontimer_StopPredefTimer : exclusive area 5
 - Gpt_Aontimer_ProcessInterrupt : exclusive area 29

- Gpt_Ftu.c:
 - Gpt_LL_Ftu_StartTimer : exclusive area 18
 - Gpt_LL_Ftu_StopTimer : exclusive area 19
 - Gpt_LL_Ftu_StartPredefTimer : exclusive area 21
 - Gpt_LL_Ftu_StopPredefTimer : exclusive area 22
 - Gpt_LL_Ftu_GetTimeElapsed : exclusive area 23

- Gpt_Rtc.c:
 - Gpt_LL_Rtc_StartTimer : exclusive area 24
 - Gpt_LL_Rtc_StopTimer : exclusive area 25
 - Gpt_LL_Rtc_EnableInterruptWithCounterUpdate : exclusive area 10
 - Gpt_LL_Rtc_EnableInterrupt : exclusive area 26
 - Gpt_LL_Rtc_DisableInterrupt : exclusive area 27
 - ISR(RTC_ISR) : exclusive area 28

- Gpt_Fcpit.c:
 - Gpt_FCPit_ProcessCommonInterrupt : exclusive area 07
 - Gpt_LL_FCPit_StopTimer : exclusive area 14
 - Gpt_LL_FCPit_EnableInterrupt : exclusive area 15
 - Gpt_LL_FCPIT_DisableInterrupt : exclusive area 16

- Gpt_Tstmp.c:
 - Gpt_LL_Tstmp_StartTimer : exclusive area 30
 - Gpt_LL_Tstmp_StopTimer : exclusive area 31
 - Gpt_LL_Tstmp_GetTimeElapsed : exclusive area 32
 - Gpt_LL_Tstmp_StartPredefTimer : exclusive area 33
 - Gpt_LL_Tstmp_StopPredefTimer : exclusive area 34

Chapter 5 Interrupt Service Routine (ISR)

Instance (FC7300)	Interrupt Name (FC7300)	IRQ Number (NVIC Interrupt ID) (FC7300)
FTU0	FTU0_IRQHandler	116
FTU1	FTU1_IRQHandler	117
FTU2	FTU2_IRQHandler	118
FTU3	FTU3_IRQHandler	119
FTU4	FTU4_IRQHandler	120
FTU5	FTU5_IRQHandler	121
FTU6	FTU6_IRQHandler	122
FTU7	FTU7_IRQHandler	123
FTU8	FTU8_IRQHandler	124
FTU9	FTU9_IRQHandler	125
FTU10	FTU10_IRQHandler	126
FTU11	FTU11_IRQHandler	127
RTC alarm interrupt	RTC_IRQHandler	65
AONTIMER0	AONTIMER_IRQHandler	71
FCPIT0	FCPIT0_IRQnHandler	63
FCPIT1	FCPIT1_IRQnHandler	64
TSTMP0	TSTMP0_IRQHandler	57
TSTMP1	TSTMP1_IRQHandler	58
TSTMP2	TSTMP2_IRQHandler	59
TSTMP3	TSTMP3_IRQHandler	60

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
Gpt_Init	GPT_E_ALREADY_INITIALIZED; GPT_E_INIT_FAILED
Gpt_DeInit	GPT_E_UNINIT; GPT_E_BUSY
Gpt_GetTimeElapsed	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL
Gpt_GetTimeRemaining	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL
Gpt_StartTimer	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL; GPT_E_PARAM_VALUE; GPT_E_BUSY
Gpt_StopTimer	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL
Gpt_EnableNotification	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL
Gpt_DisableNotification	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL
Gpt_SetMode	GPT_E_UNINIT; GPT_E_PARAM_MODE
Gpt_DisableWakeup	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL
Gpt_EnableWakeup	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL
Gpt_CheckWakeup	GPT_E_UNINIT;
Gpt_ChangeNextTimeoutValue	GPT_E_UNINIT; GPT_E_PARAM_CHANNEL; GPT_E_BUSY; GPT_E_PARAM_VALUE
Gpt_GetPredefTimerValue	GPT_E_UNINIT; GPT_E_PARAM_POINTER; GPT_E_PARAM_PREDEF_TIMER
Gpt_GetVersionInfo	GPT_E_PARAM_POINTER
Gpt_StartGlobalTime	GPT_E_UNINIT GPT_E_PARAM_GROUP
Gpt_StopGlobalTime	GPT_E_UNINIT GPT_E_PARAM_GROUP

6.2 Dem

None

Chapter 7 Function Calls

7.1 Function Calls during Startup

The API needs to be called is `GPT_Init(ConfigPtr);`

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

None

Chapter 8 Other Requirements

8.1 Notification, Callback, Callout

- Notification

If the user configures notifications and the value is not "NULL_PTR" or "NULL", an extern declaration will generate in Gpt_Cfg.h. User need implement the notification in any file which include Gpt_Cfg.h.

- Callback

None

- Callout

None

8.2 Macros

Please check various definitions available in Common module's include file Mcal.h for details.

- If OS is not used:

AUTOSAR_OS_NOT_USED needs to be defined.

In this case, If USE_SW_VECTOR_MODE is not defined, the user needs to map the interrupt vector table function to ISR function, for example:

```
extern ISR(FTU_0_ISR);

void FTU0_IRQHandler(void)

{

    FTU_0_ISR();

}
```

- If OS is used:

Do not define AUTOSAR_OS_NOT_USED.

Chapter 9 Integration Steps

- 1) Configure the GPT module and generate configuration files (please refer to Building chapter for details).
- 2) Configure appropriate memory sections in linker file or other (please refer to Memory chapter for details).
- 3) Map interrupt notification to their vector locations (please refer to chapter ISR for details).
- 4) Build the GPT module with dependent modules.